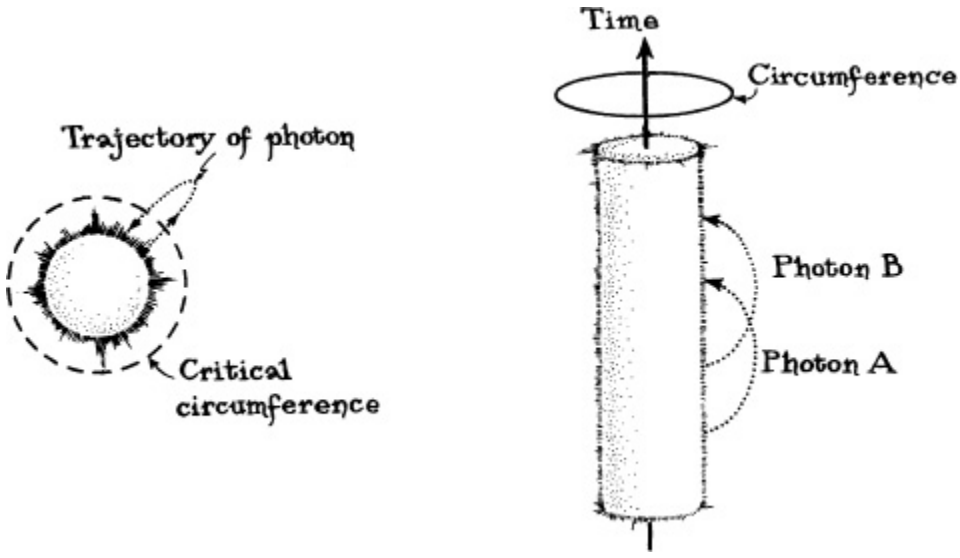
The left half of Figure 6.8 (a space diagram, not a spacetime diagram) depicts such a star inside its critical circumference, and depicts the spatial trajectory of a photon (light corpuscle) emitted from the star's surface nearly vertically (radially). The outflying photon, like a thrown rock, is slowed by the pull of the star's gravity, it draws to a halt, and it then falls back into the star.

The right half of the figure depicts in a spacetime diagram the motions of two such photons. Plotted upward is Newton's universal time; plotted outward, his absolute space. With the passage of time, the circular star sweeps out the vertical cylinder; at any moment of time (horizontal slice through the diagram) the star is described by the same circle as in the left picture. As time passes, photon A flies out and then falls back into the star, and photon B, emitted a little later, does the same.



6.8 The predictions from Newton's laws of physics for the motion of light corpuscles (photons) emitted by a star that is inside its critical circumference. *Left:* a spatial diagram (similar to Figure 3.1). *Right:* a spacetime diagram.

It is instructive to compare this (incorrect) Newtonian version of a star inside its critical circumference and the photons it emits with the (correct) relativistic version, Figure 6.7. The comparison shows two profound differences between the predictions of Newton's laws and those of Einstein:

1. Newton's laws (Figure 6.8) permit a star smaller than the critical circumference to live a happy, non-imploding life, with its gravitational squeeze forever counterbalanced by its internal